– Abstract Algebra

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Work Sheet Stuff

5) We can assume associativity

4)

$\circ$	e	$r(120°)$	$r(240°)$	s	s'	s''
e						
$r(120°)$						
$r(240°)$						
s						
s'						
s''						

Play the game with a square! ☺

→ Check if it is a group. Check if it's Abelian.

r & s are "generators" of this group because all compositions of 120° rotations or vertical, left diagonal, or right diagonal reflections can be represented by either r, s, or id. composed with eachother (see in-class worksheet).

Lesson: Highlights / Punch Lines from our work sheet:

- the symmetries of a triangle form a group.
- The group opperation is function composition, and the group has six elements.

If we lable ${}_3\overset{1}{\Delta}_2 \xrightarrow{r} {}_2\overset{3}{\Delta}_1$ and ${}_3\overset{1}{\Delta}_2 \xrightarrow{s} {}_2\overset{1}{\Delta}_3$. The elements are

$\{e = id., r, r \circ r = r^2, s, rs, r^2s\}$.

This group is called the Dihedral Group on 3 verticies. We denote it by D_3.

(Some authors denote it as D_6 for number of elements is 6).

Generators & Relations

$$D_3 = \langle \underbrace{r, s}_{\text{generators}} : r^3 = s^2 = e \text{ and } rs = sr^2 \rangle.$$

↳ "subject to the relations"

This means D_3 is generated by symbols r & s subject to the relations $r^3 = s^2 = e$ and $rs = sr^2$.

D_3 is not Abelian (it is not commutative). In particular, $rs \neq sr$.

Back to the text book/ our regularly scheduled programming:

Thm. 2.4. The Socks - Shoes Property

Let $a, b \in G$, where G is a group. The inverse of ab is the inverse of b group opperationed $(*)$ with the inverse of a.

i.e. ab's invere is $(ab)^{-1} = b^{-1}a^{-1}$, where -1 is a superscript to denote inverse.

It's called the socks shoes property because if b = putting on socks and a = putting on shoes

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then, to undo this action, we must 1st take off our shoes and then take off our socks (i.e. $\underset{1st}{\underline{b^{-1}}}\ \underset{2nd}{\underline{a^{-1}}}$).

<u>Proof</u>: We compute $b^{-1}\underline{a^{-1}\cdot ab} = b^{-1}\underline{e}\,b$

$= b^{-1}b = e$. $ab\,\underline{b^{-1}a^{-1}} = a\underline{e}a^{-1} = aa^{-1} = e$.

Thus $b^{-1}a^{-1}$ is an inverse to ab.

However, since the inverse to any element is <u>unique</u>, then $b^{-1}a^{-1} = (ab)^{-1}$. ∎